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Agentic AI in Production: Designing Autonomous Multi-Agent Systems with Guardrails (2026 Guide)

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Dewasheesh Rana Follow

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> *Agentic AI is not about letting LLMs "think more."*
> *It is about letting systems act — safely, deterministically, and within boundaries.*

Over the last year, *Agentic AI* has become one of the most overused terms in Generative AI.
Most articles stop at demos: planners, tool calls, and impressive traces — but very few address the **hard part**:

> **How do you run agentic systems in production without losing control?**

This article explains **what Agentic AI really is**, **why most implementations fail**, and **how to design production-grade multi-agent systems with guardrails.**

1. What Agentic AI Actually Means (Not the Marketing Version)

Agentic AI does **not** mean:

- Longer prompts
- "Chain-of-thought everywhere"
- Giving the model more autonomy blindly

Agentic AI means:

> A system where an LLM can **plan, decide, act, observe outcomes, and adjust**, using tools and memory — *within explicitly defined constraints*.

A true agent has:

- **Goals**
- **Decision-making authority**
- **Access to actions (tools)**
- **Feedback loops**
- **Stopping conditions**

If your system only calls tools sequentially in a fixed flow, it is **not agentic** — it is a workflow.

2. Agent vs Tool vs Workflow (Critical Mental Model)

Most confusion disappears once this distinction is clear.

◆ Tool

- A deterministic function
- No decision-making

- Example: `search_docs()`, `send_email()`

◆ Workflow

- Predefined execution path
- Conditional logic, but static
- Example: traditional RAG pipeline

◆ Agent

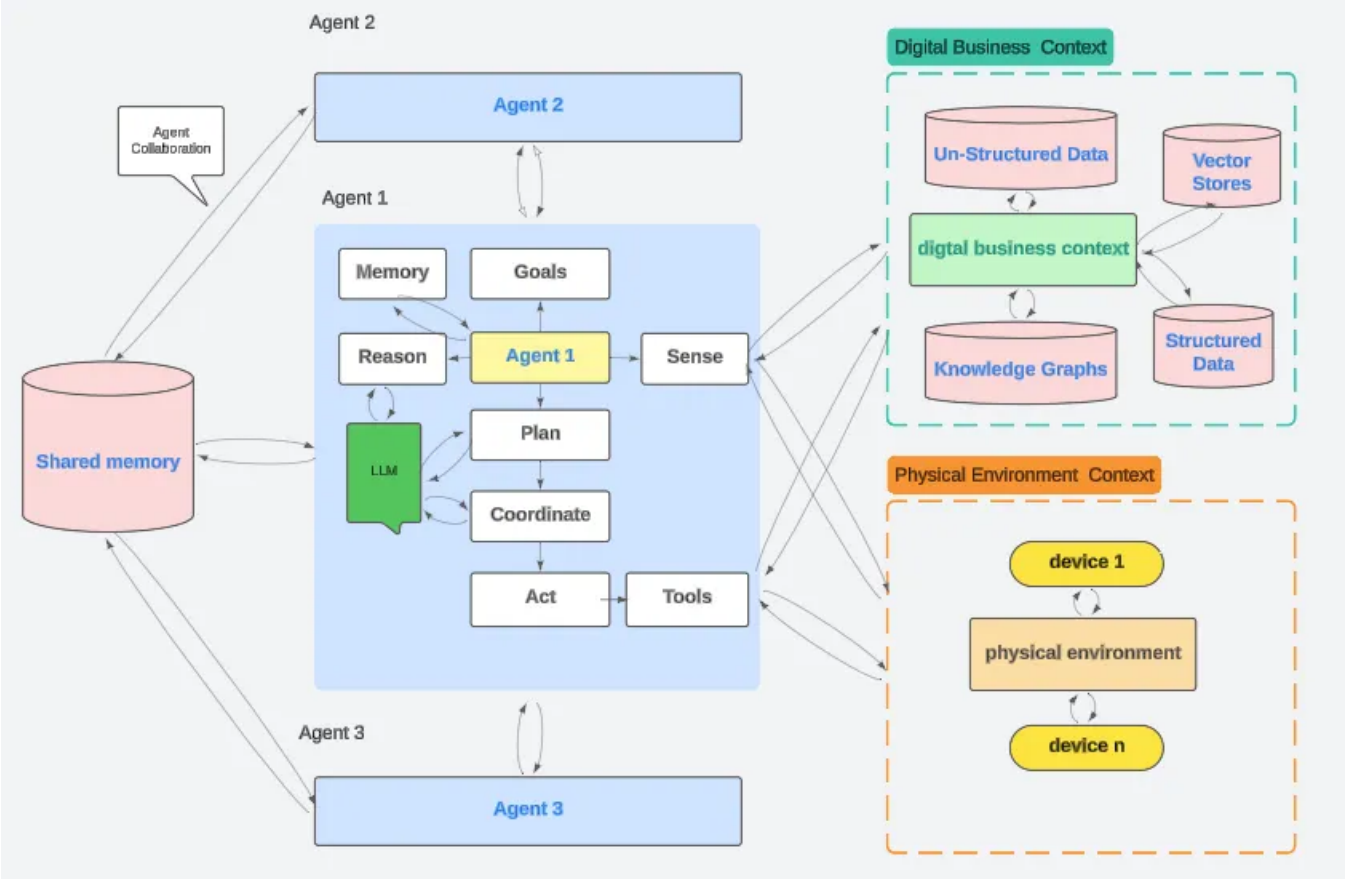
- Decides **what to do next**
- Can choose tools, re-plan, or stop
- Operates under constraints

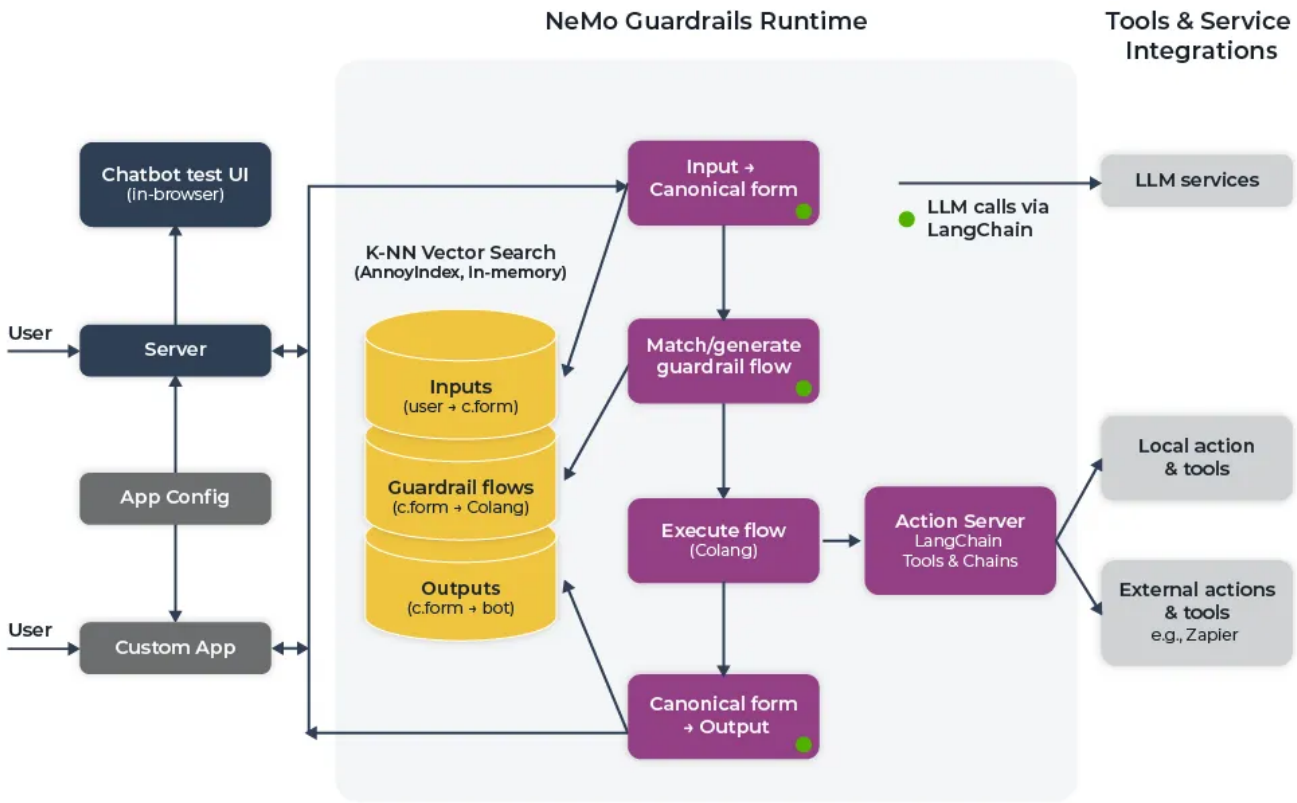
> **Production Agentic AI = Agents + Tools + Controlled Workflows**

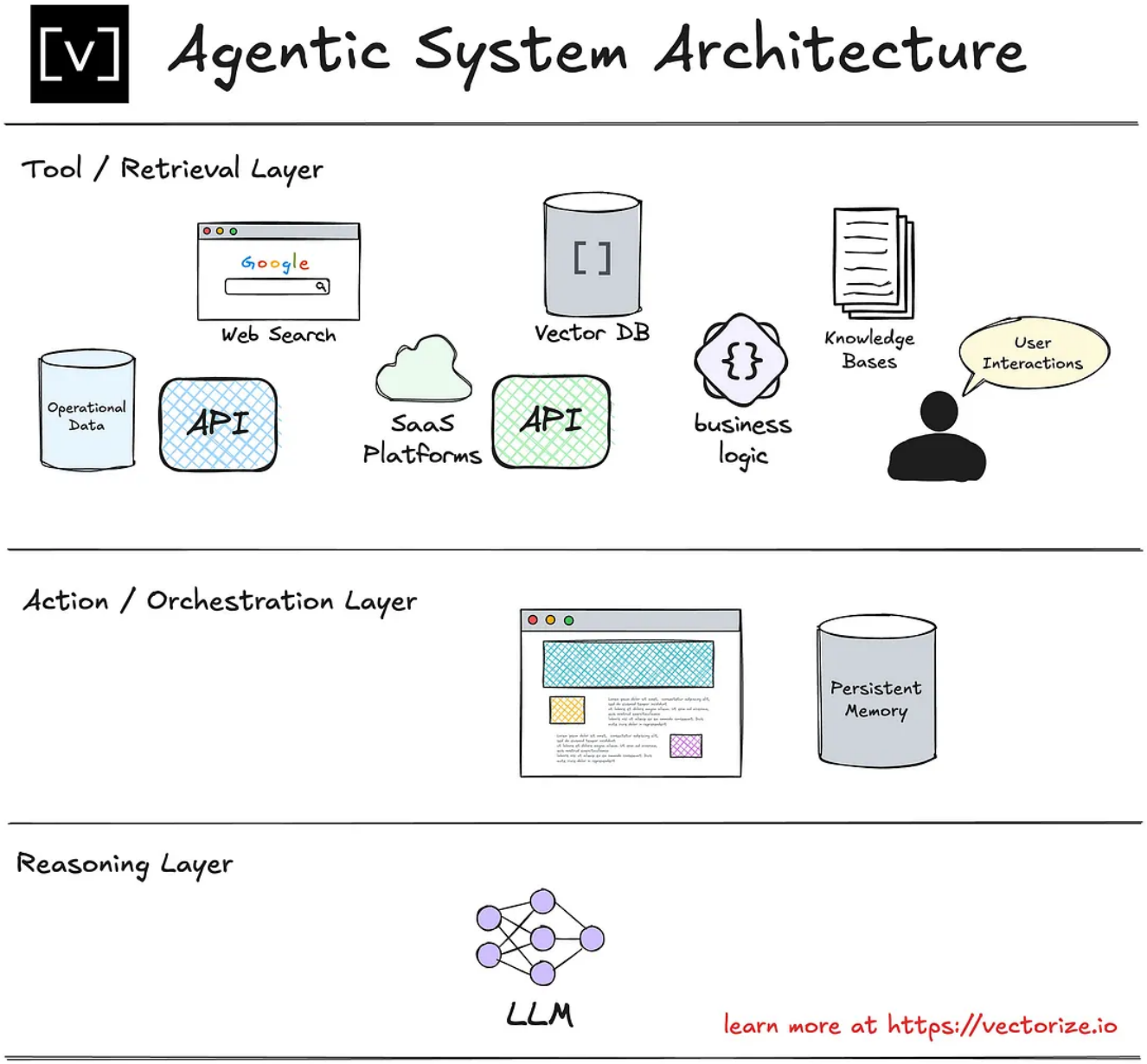
Never deploy a *free-roaming agent*.

3. The Production Multi-Agent Architecture

A single agent does not scale well.
Real systems use **specialized agents** with narrow responsibilities.


Agent 2
Agent Collaboration
Agent 2
Agent 1
Memory
Goals
Reason
Agent 1
Sense
LLM
Plan
Coordinate
Act
Tools
Shared memory
Agent 3
Agent 3
Digital Business Context
Un-Structured Data
Vector Stores
digtal business context
Knowledge Graphs
Structured Data
Physical Environment Context
device 1
physical environment
device n


NeMo Guardrails Runtime
Tools & Service Integrations
Chatbot test UI
(in-browser)
User
Server
App Config
User
Custom App
K-NN Vector Search
(AnnoyIndex, in-memory)
Inputs
(user → c.form)
Guardrail flows
(c.form → Colang)
Outputs
(c.form → bot)
Input → Canonical form
Match/generate guardrail flow
Execute flow
(Colang)
Canonical form → Output
LLM calls via LangChain
LLM services
Action Server
LangChain
Tools & Chains
Local action & tools
External actions & tools
e.g., Zapier



Recommended Core Agents

1. **Planner Agent**

- Breaks the user goal into steps
- Decides which agent should act next

2. Executor Agent

- Performs tool calls

- No creative freedom

3. Critic / Validator Agent

- Reviews outputs
- Detects hallucinations or unsafe actions

4. Memory Agent

- Manages short-term + long-term memory
- Prevents context explosion

5. Supervisor (Optional but Powerful)

- Enforces budgets, step limits, and policies
- Can terminate execution

> This separation is the **single biggest factor** between toy demos and production systems.

4. Why Most Agentic AI Systems Fail in Production

Let's be honest about failure modes.

❌ Infinite Reasoning Loops

Agents keep "thinking" because no stop condition exists.

Fix:

- Max steps
- Goal satisfaction checks
- Forced termination

❌ Tool Abuse

Agents call tools repeatedly or incorrectly.

Fix:

- Tool permission matrix
- Rate limits per tool
- Tool schemas with strict validation

❌ Non-Deterministic Behavior

Same input → different actions → unpredictable system.

Fix:

- Low temperature for planning
- Deterministic tool calls
- Separate reasoning from execution

❌ No Observability

You can't debug what you can't see.

Fix:

- Log every decision
- Trace every tool call

- Store intermediate plans

5. Guardrails: The Non-Negotiable Layer

Guardrails are not optional. They are the system.

Mandatory Guardrails for Agentic AI

1. Action Boundaries

Agents should only access **explicitly allowed tools.**

```
Planner Agent → allowed: reasoning only
Executor Agent → allowed: tools A, B
Critic Agent → allowed: validation only
```

2. Budget & Step Limits

- Max reasoning steps
- Max tool calls
- Max cost per request

> If an agent exceeds limits → terminate safely.

3. Human-in-the-Loop (Selective)

Not everywhere — only where risk is high:

- Financial actions

- Data deletion
- External communication

4. Memory Guardrails

- What can be stored
- How long it lives
- Who can access it

6. Agentic RAG: The Most Practical Use Case

The best real-world use of Agentic AI today is **Agentic RAG.**

Traditional RAG

- Always retrieves
- Always answers

Agentic RAG

- Decides **when to retrieve**
- Decides **what to retrieve**
- Decides **when to stop**

Flow:

```
User Query →
Intent Agent →
   ↳ Retrieve (if needed)
   ↳ Reason Only (if not)
```

```
    ↳ Tool Call (if required)
```

This:

- Reduces cost
- Improves accuracy
- Avoids unnecessary retrieval

7. Minimal Production-Safe Agent Loop (Pseudo-Code)

```
while steps < MAX_STEPS:
    plan = planner.decide(state)

    if plan.action == "STOP":
        break

    if not supervisor.is_allowed(plan):
        raise SafetyException

    result = executor.execute(plan)
    state.update(result)

    if critic.rejects(state):
        rollback()
        break
```

Notice:

- No free loops
- No unchecked tool calls
- Explicit stopping logic

8. When You Should NOT Use Agentic AI

Agentic AI is powerful — but dangerous when misused.

❌ Do NOT use agents when:

- The task is deterministic
- Latency must be ultra-low
- Regulatory compliance is strict
- A simple workflow works fine

> **Agentic AI should reduce human effort — not system reliability.**

9. The Future: Controlled Autonomy, Not Full Autonomy

The industry is converging on a clear direction:

- Fewer "general agents"
- More **specialized, constrained agents**
- Strong supervision layers
- Observability-first design

The winners will not be the systems that *think the most* —
but the ones that **fail safely**.

Final Takeaway

> **Production Agentic AI is a systems engineering problem, not a prompting problem.**

If your design does not include:

- Clear agent roles
- Hard guardrails
- Deterministic execution paths
- Observability and rollback

You don't have an agentic system —
you have a **risk amplifier**.

Example:

Agentic AI in Practice: Building a Production-Safe Multi-Agent System with LangGraph and CrewAI

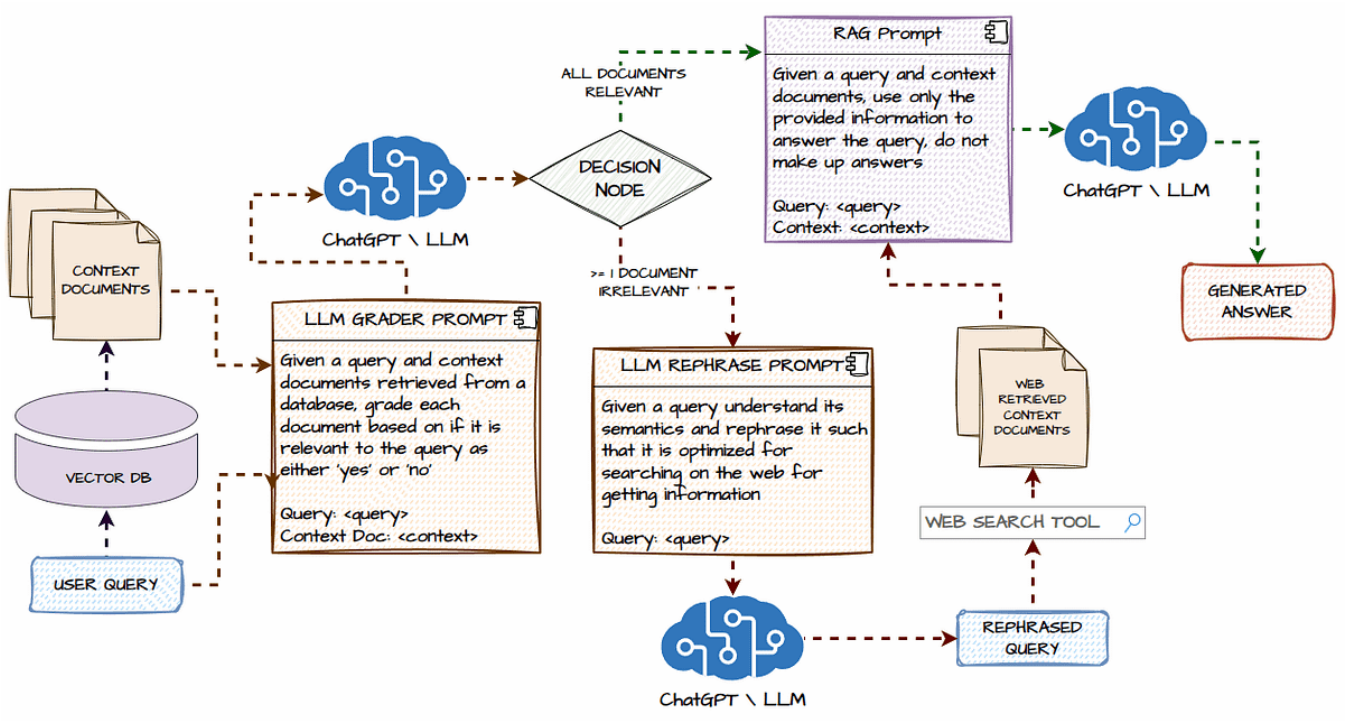
> *Agentic AI becomes real only when agents can act — and stop — safely.*

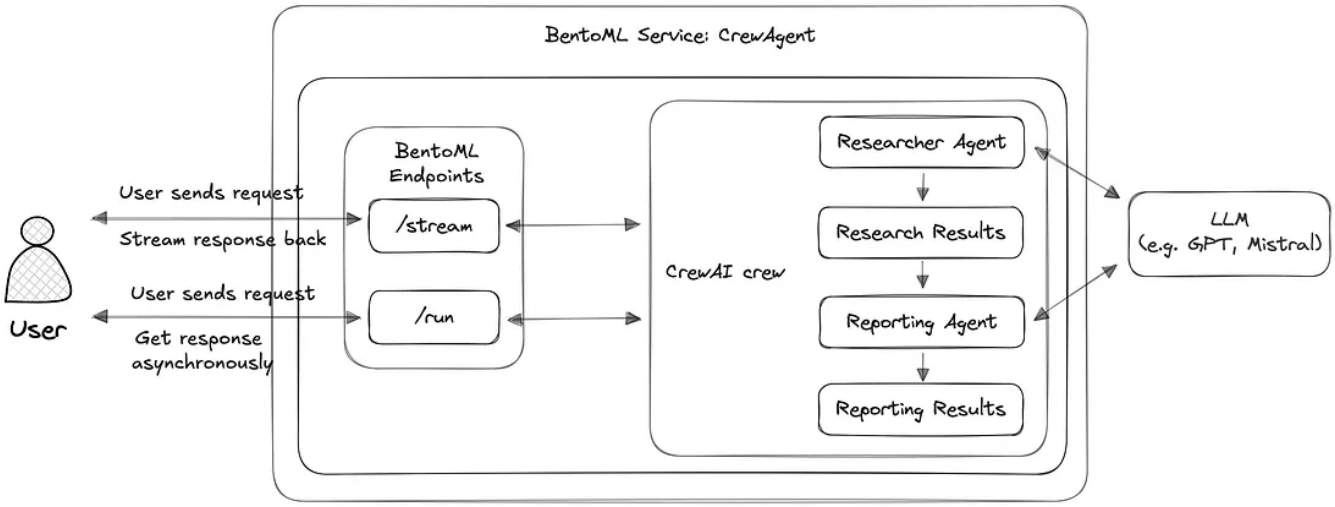
In the above section, we discussed **why most agentic AI systems fail in production** and how **guardrails** are non-negotiable.
This follow-up goes one level deeper:

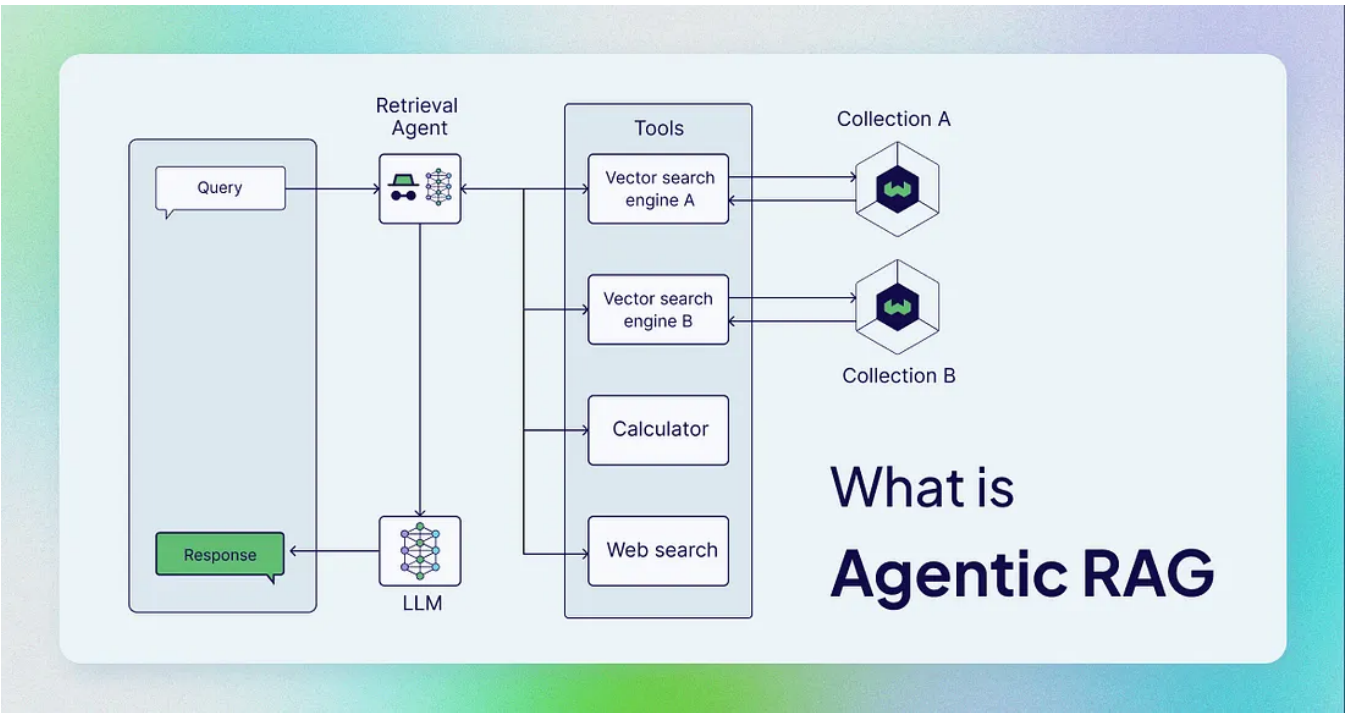
> **We will build a real, minimal, production-safe multi-agent system using LangGraph and CrewAI.**

No toy demos.
No infinite loops.
No "let the agent decide everything" anti-patterns.

1. Target Architecture (What We're Actually Building)







Agents in Scope

Agent	Responsibility
Intent Agent	Decide *what kind of task this is*
RAG Agent	Retrieve knowledge only if needed
Tool Agent	Execute deterministic actions
Critic Agent	Validate output
Supervisor	Enforce limits & stop conditions

Key rule:

> No agent can directly call another agent — all routing is explicit.

2. Why LangGraph + CrewAI Together?

LangGraph

- Deterministic control flow

- State machine for agents
- Explicit stop conditions

CrewAI

- Role-based agents
- Clear separation of responsibility
- Easy reasoning + tool abstraction

> **LangGraph controls *flow*. CrewAI controls *thinking*.**

This separation is crucial in production.

3. Project Structure (Production-Friendly)

```
agentic_system/
├── graph.py
├── agents/
│   ├── intent_agent.py
│   ├── rag_agent.py
│   ├── tool_agent.py
│   ├── critic_agent.py
├── supervisor.py
├── tools.py
└── main.py
```

4. Step 1 — Define the Shared State (LangGraph)

```
from typing import TypedDict, Optional

class AgentState(TypedDict):
    user_input: str
    intent: Optional[str]
    context: Optional[str]
    output: Optional[str]
    steps: int
```

Why this matters

- Every agent sees the same state
- No hidden memory
- Easy debugging & tracing

5. Step 2 — Intent Agent (CrewAI)

```
from crewai import Agent

intent_agent = Agent(
    role="Intent Classifier",
    goal="Identify whether the user needs retrieval, tools, or reasoning",
    backstory="Expert at routing tasks safely",
    verbose=False
)

def run_intent_agent(state: AgentState):
    prompt = f"""
    Classify intent for the input:
    '{state["user_input"]}'
    Return one word: RAG | TOOL | REASON
    """
    intent = intent_agent.run(prompt).strip()
```

```
    state["intent"] = intent
    state["steps"] += 1
    return state
```

✅ No tool access
✅ Low temperature recommended
✅ Single responsibility

6. Step 3 — RAG Agent (Controlled Retrieval)

```
from crewai import Agent
rag_agent = Agent(
    role="Knowledge Retrieval Agent",
    goal="Retrieve only relevant information",
    backstory="Precision-focused research agent",
    verbose=False
)

def run_rag_agent(state: AgentState):
    if state["intent"] != "RAG":
        return state

    context = rag_agent.run(
        f"Retrieve concise context for: {state['user_input']}"
    )
    state["context"] = context
    state["steps"] += 1
    return state
```

⚠️ Retrieval happens **only if intent says so**
⚠️ This avoids unnecessary vector DB calls

7. Step 4 — Tool Agent (Zero Creativity)

```
from crewai import Agent
def calculator_tool(expression: str) -> str:
    return str(eval(expression))

tool_agent = Agent(
    role="Tool Executor",
    goal="Execute tools exactly as instructed",
    backstory="No creativity. Deterministic execution only.",
    tools=[calculator_tool],
    verbose=False
)

def run_tool_agent(state: AgentState):
    if state["intent"] != "TOOL":
        return state

    result = tool_agent.run(state["user_input"])
    state["output"] = result
    state["steps"] += 1
    return state
```

🚫 Tool agent **never reasons**

🚫 Tool schemas must be strict

8. Step 5 — Critic Agent (Hallucination Guard)

```
from crewai import Agent
critic_agent = Agent(
    role="Output Validator",
    goal="Validate correctness and safety",
    backstory="Extremely strict reviewer",
    verbose=False
)
```

```
def run_critic_agent(state: AgentState):
    review = critic_agent.run(
        f"Validate the output: {state.get('output')}"
    )
    if "invalid" in review.lower():
        state["output"] = "Unable to safely answer."
    state["steps"] += 1
    return state
```

This agent:

- Prevents hallucinated answers
- Enables rollback strategies
- Is **mandatory** in production

9. Step 6 — Supervisor (The Kill Switch)

```
MAX_STEPS = 6

def supervisor(state: AgentState):
    if state["steps"] >= MAX_STEPS:
        state["output"] = "Stopped due to step limit."
        return False
    return True
```

> **If you don't have a kill switch, you don't have an agentic system.**

10. Step 7 — Wiring Everything with LangGraph

```
from langgraph.graph import StateGraph

graph = StateGraph(AgentState)

graph.add_node("intent", run_intent_agent)
graph.add_node("rag", run_rag_agent)
graph.add_node("tool", run_tool_agent)
graph.add_node("critic", run_critic_agent)

graph.set_entry_point("intent")

graph.add_edge("intent", "rag")
graph.add_edge("rag", "tool")
graph.add_edge("tool", "critic")

graph.set_conditional_edges(
    "critic",
    supervisor,
    {True: "intent", False: None}
)

agentic_app = graph.compile()
```

✅ Explicit transitions
✅ Deterministic flow
✅ Controlled looping

11. Final Execution

```
result = agentic_app.invoke({
    "user_input": "What is 25 * 42?",
    "steps": 0
})
print(result["output"])
```

12. Why This Design Works in Production

✓ No infinite loops
✓ No uncontrolled tools
✓ Clear observability
✓ Predictable cost
✓ Easy debugging

This is **how real agentic systems should be built.**

13. What Most Tutorials Get Wrong

❌ Single “super agent”
❌ Unlimited autonomy
❌ No validation agent
❌ No stop condition
❌ No state transparency

> **Autonomy without control is not intelligence — it’s risk.**

Final Takeaway

> **LangGraph gives you structure.**
> **CrewAI gives you reasoning.**
> **Guardrails give you safety.**

Combine all three — or don’t ship agentic AI at all.

Agentic Ai Architecture

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Written by Dewasheesh Rana

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Dewasheesh Rana is an AI/ML Architect with 14+ yrs experience building enterprise systems using LLMs, RAG, MLOps, and backend engineering.

Responses (6)

Tárik César

What are your thoughts?

Seamus Galvin
1 day ago

Nice article - thank you!

Reply

Abdel Otmani
3 days ago

I wish I saw it earlier before we experienced it by ourselves

Do you have a some recommendation on how to create guardrails and which one to apply depending on the agent?

Reply

Tim Dentry
5 days ago

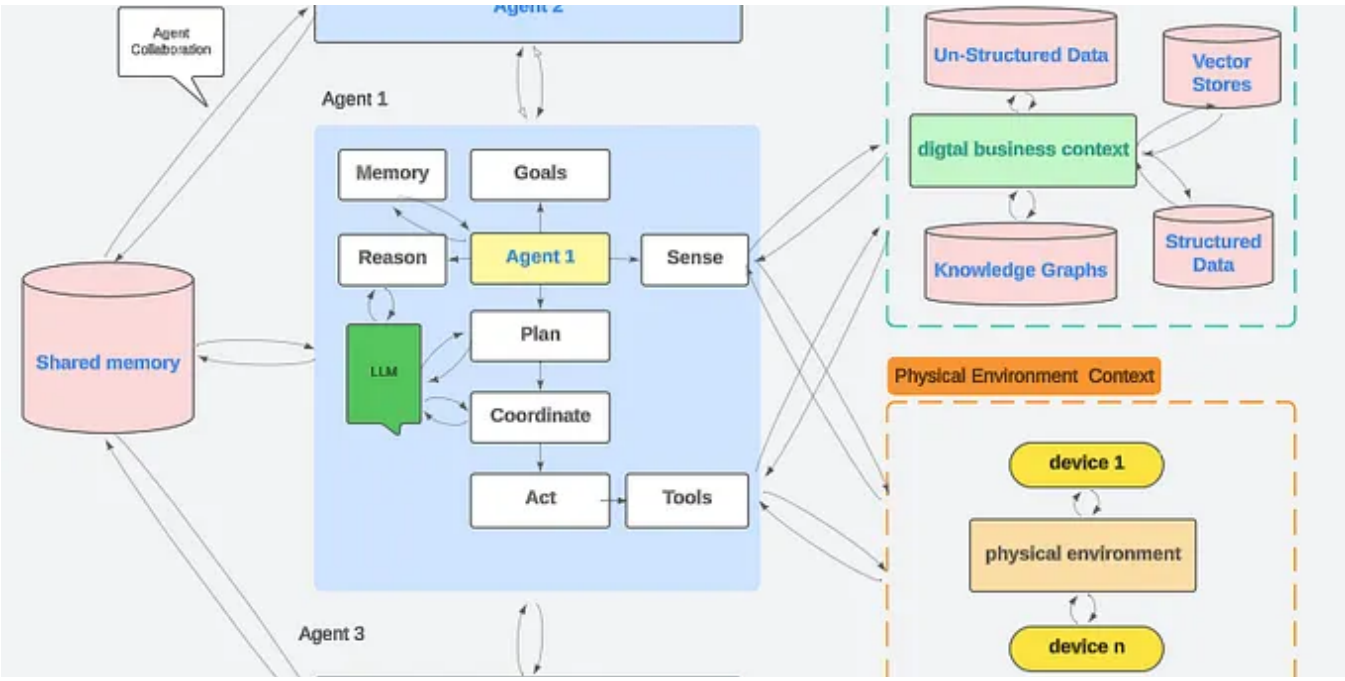
Actually

What is the diagramming tool you are using? The script makes it look like handwriting instead of boring old text :-)

1 reply Reply

See all responses

More from Dewasheesh Rana



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Agentic AI Design Patterns(2026 Ed

The Definitive Architect's Guide (Jan 2026)

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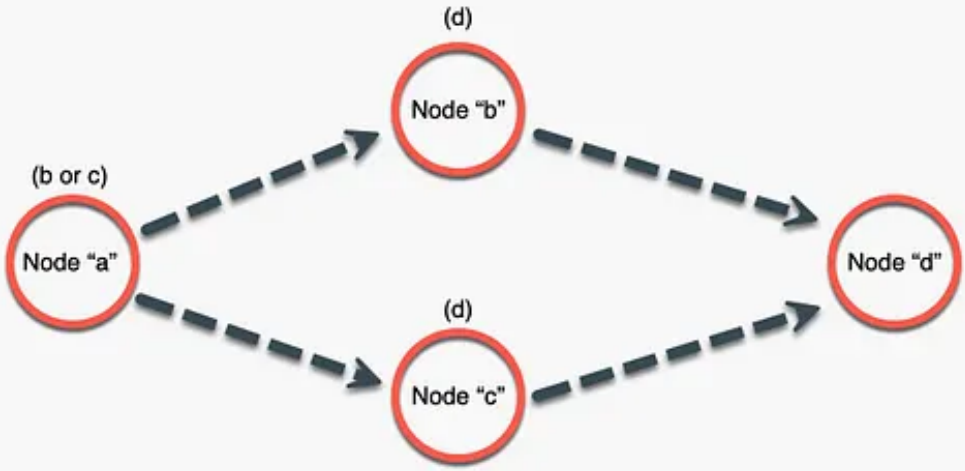

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LangGraph Explained (2026 Edition)

From Zero → Engineer → Architect → Production

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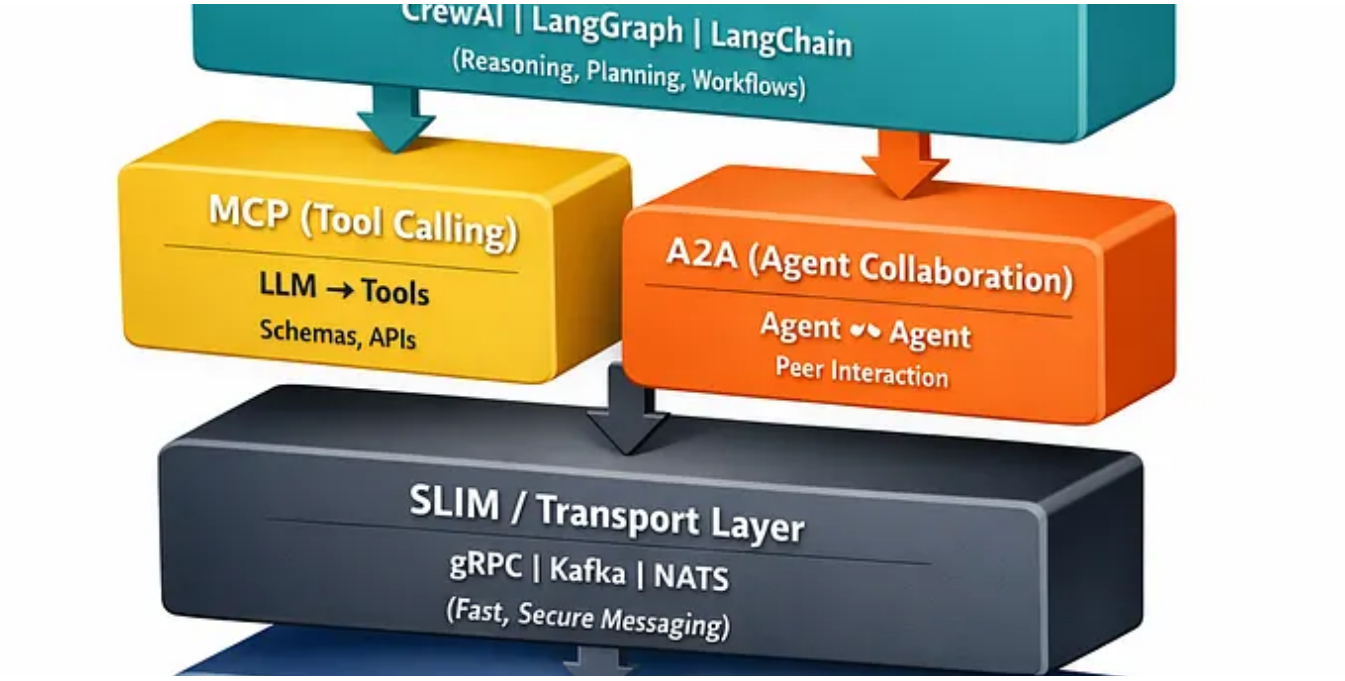

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Agent Communication Protocols Explained (MCP, A2A, SLIM, ACP)

A Complete, Practical Guide for Students, Engineers, and Architects

Dec 30, 2025 64 1

g code, we establish **non-negotiable boundaries:**

Responsibility	What it must NOT do
Fast, efficient inference	Workflow logic
State & control flow	Model optimization
Probabilistic reasoning	System decisions
Deterministic execution	Free-form reasoning

tion is the foundation of reliability.

Dewasheesh Rana

Building High-Performance Agentic AI with vLLM and LangGraph

A Canonical, Production-Grade Reference (No Hand-Waving)

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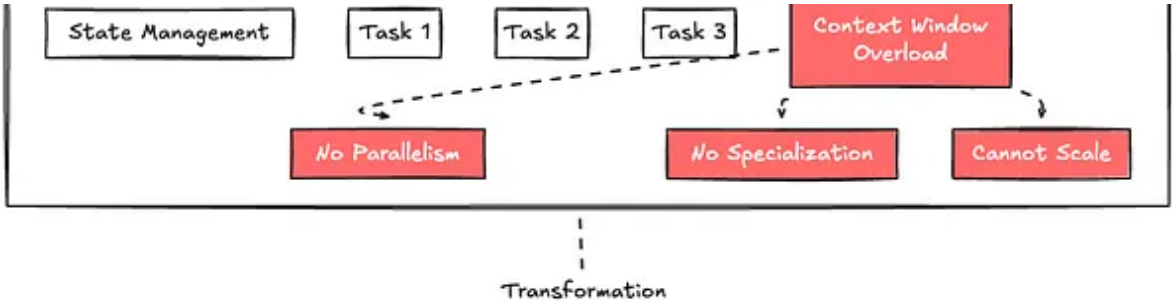

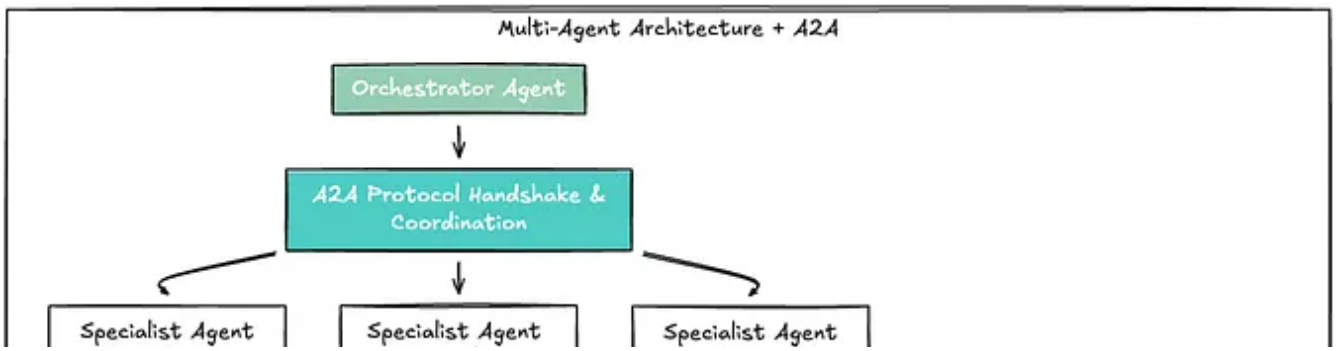

Yusuf Baykaloğlu

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A Deep Dive into Agent Architecture, Workflows, and Real-World Implementation

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In AI Software Engineer by Joe Njenga

Anthropic Just Released Claude Code Course (And I Earned My Certificate)

Anthropic just launched their Claude Code in Action course, and I've just passed — how about you?

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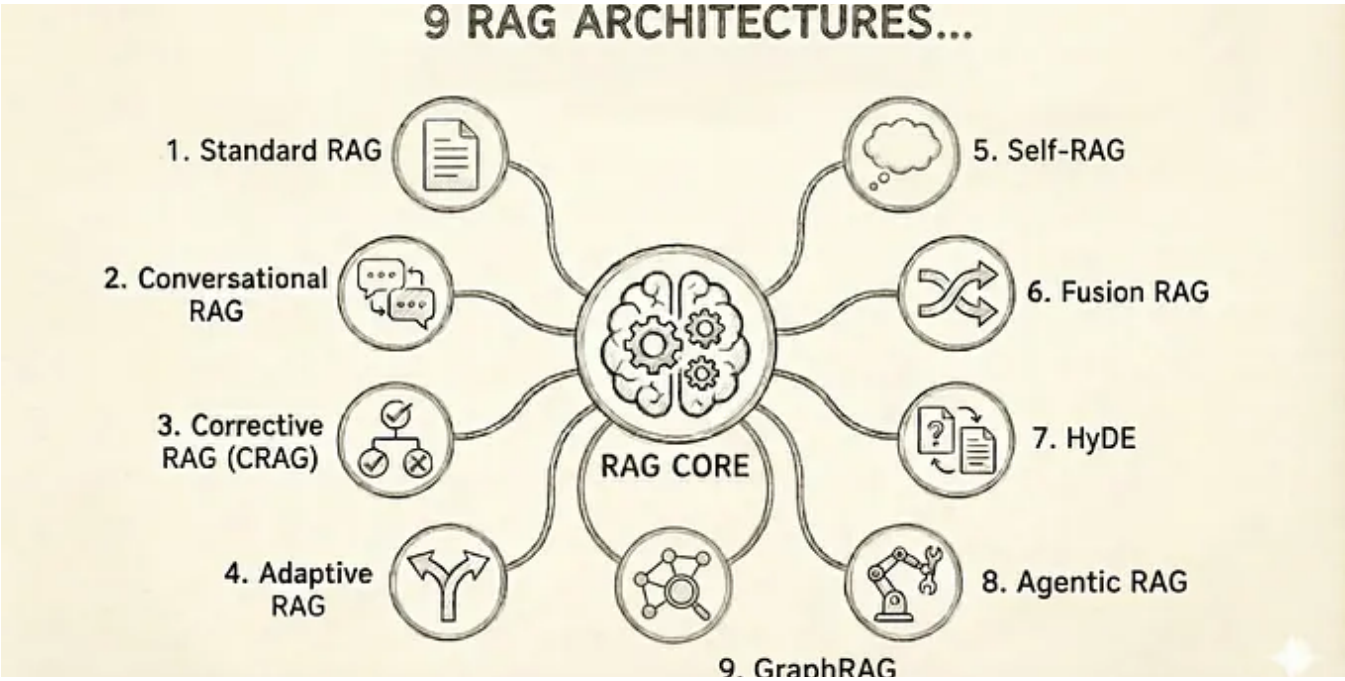

In Towards AI by Divy Yadav

9 RAG Architectures Every AI Developer Must Know: A Complete Guide with Examples

Architectures beyond Naive Rag to build reliable production AI Systems

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Reza Rezvani

I Tested Every Major Claude Opus 4.6 Feature — Here's What Actually Matters

After 24 hours of real testing across my daily workflow, here's my honest calibration on the new Anthropic's Frontier Model

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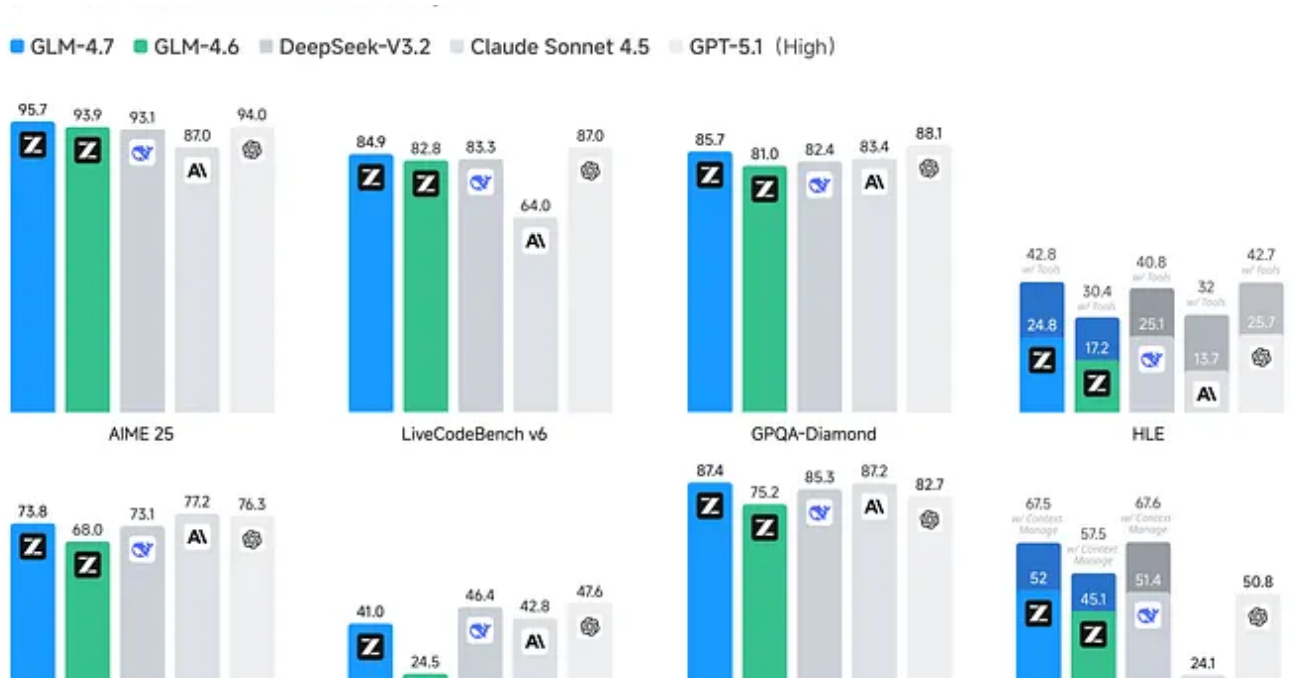

Agent Native

Local LLMs That Can Replace Claude Code

Small team of engineers can easily burn >$2K/mo on Anthropic's Claude Code (Sonnet/Opus 4.5). As budgets are tight, you might be wondering...

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